

Due on Tuesday, February 28th.

If pictures are needed/relevant, please provide them with your solutions.

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This problem set is about the relationships between electric field, electric potential and electric charge density.

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1. **[4 pts.]** Using cartesian coordinates, show that the curl of the gradient of a scalar field is zero, i.e.,  $\nabla \times (\nabla f) = 0$  for any scalar field  $f(x, y, z)$ .

[Note: this means: a vector field that can be expressed as a gradient of a scalar field, must have zero curl. Thus the electric field, which is the gradient of  $-V$ , must have zero curl:  $\nabla \times \mathbf{E} = 0$ .]

2. The electric potential in some region of space is given by

$$V(x, y, z) = \frac{C}{\sqrt{x^2 + y^2 + z^2}}$$

or, in spherical coordinates,  $V(r) = C/r$ . Here  $C$  is a positive constant.

- (a) **[5 pts.]** Calculate an expression for the electric field in this region, using cartesian coordinates.
  - (b) **[4 pts.]** Noting that  $x\hat{i} + y\hat{j} + z\hat{k} = \mathbf{r} = r\hat{r}$ , express the expression for  $\mathbf{E}$  in spherical coordinates. Here  $\hat{r}$  is the radial unit vector of spherical coordinates, sometimes also denoted as  $\mathbf{e}_r$  or  $\hat{e}_r$ .
  - (c) **[4 pts.]** Look up the expression for the gradient operator in spherical coordinates. Re-derive the expression for  $\mathbf{E}$ , now directly in spherical coordinates.
  - (d) **[5 pts.]** Look up the expression for the Laplacian operator in spherical coordinates. By calculating the Laplacian of  $V$ , find out the charge density  $\rho$  in this region.
  - (e) **[5 pts.]** Calculate the Laplacian of  $V$ , and hence the charge density  $\rho$ , using cartesian coordinates. (This is a cumbersome calculation.)
3. **[5 pts.]** Show that the vector field  $\mathbf{A} = y\hat{i} - x\hat{j}$  cannot be a static electric field, because it's curl is nonzero. Which direction does  $\nabla \times \mathbf{A}$  point in?

4. The electric field is claimed to be  $\mathbf{E} = x\hat{i} + y\hat{j}$  in some region of space.
- (a) **[5 pts.]** Show that this satisfies the requirement of zero curl.
  - (b) **[5 pts.]** Using Maxwell's first equation, find the electric charge density  $\rho$  in this region of space. How does  $\rho$  vary in space?
  - (c) **[3 pts.]** Calculate the difference of electric potential between the points  $(1, 5, -1)$  and  $(1, 5, +1)$ .
  - (d) **[5 pts.]** Calculate the difference of electric potential between the points  $(0, 5, 0)$  and  $(2, 5, 0)$ .